

Failure Mode and Effects Analysis (FMEA)

Document ID:	FMEA-CKMB-001
Product:	NEXUS CK MB (30421)
Version:	2.0
Effective Date:	2023-01-15
Author:	J. Martin - Quality Engineering
Approved By:	P. Dubois - Quality Director

1. Purpose

This FMEA evaluates potential failure modes in the manufacturing process of NEXUS CK MB (Reference 30421) immunoassay kits. The analysis covers all critical process steps from SPR cone receipt through final packaging, with particular attention to calibration-related risks.

2. Scope

Applicable to the complete NEXUS CK MB manufacturing process at the Marcy-l'Etoile production facility. Process steps: SPR Receipt, Strip Assembly, Kit Assembly, Calibration, QC Testing, Labeling, Packaging.

3. FMEA Risk Assessment

Process Step	Failure Mode	Effect	S	Cause	O	Controls	D	RPN
SPR_Receipt	SPR coating variation	Signal out of specification	8	Supplier lot-to-lot variation	3	Incoming inspection per SOP-SPR-001	4	96
Calibration	Calibration drift	Invalid calibration (B18)	9	Equipment aging/wear	2	Quarterly PM, trend monitoring	5	90
Kit_Assembly	Wrong component	Incorrect test results	9	Labeling error	1	Visual inspection, barcode scan	3	27
Strip_Assembly	Reagent deposition error	Reduced assay sensitivity	7	Dispenser nozzle clog	2	Daily nozzle verification, flow check	3	42

Process Step	Failure Mode	Effect	S	Cause	O	Controls	D	RPN
QC_Testing	Incorrect sample handling	False QC result	6	Operator training gap	2	Qualification programme, double-check procedure	3	36
Labeling	Mislabeling of kit	Product traceability loss	7	Printer malfunction	1	Barcode verification at labeling station	2	14
Packaging	Incomplete kit content	Customer complaint	6	Missing component during assembly	2	Weight check, visual inspection checklist	3	36
SPR_Receipt	Expired SPR cones used	Degraded antibody performance	8	FIFO not followed in storage	1	ERP lot tracking, FIFO storage labels	2	16

4. Risk Priority Summary

Critical threshold: RPN >= 80. Two failure modes exceed the critical threshold and require enhanced monitoring:

- **SPR coating variation (RPN=96):** Incoming inspection enhanced with antibody density measurement per SOP-SPR-001.
- **Calibration drift (RPN=90):** Quarterly preventive maintenance and continuous trend monitoring of S1/S2 signals.

5. Revision History

Version	Date	Author	Changes
1.0	2021-06-01	J. Martin	Initial release
1.5	2022-03-15	J. Martin	Updated RPN for calibration drift after INV-2022-0045
2.0	2023-01-15	J. Martin	Added SPR coating variation risk; updated controls